ACTIVE LONGEVITY INSTITUTE

Institute Collaborator Portal

Final Proposal

Foundation Release (Version 1.0)

Prepared in response to the Final Development Brief, August 2026

Submitted by

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Confidential — prepared exclusively for the Active Longevity Institute

1. Executive Summary

This proposal responds directly to the Institute Collaborator Portal — Final Development Brief (Foundation Release, Version 1.0) and to the seven questions set out in Section 15.

In summary:

- I accept the Foundation Release scope as defined, with one addition and three simplifications, each explained and each cost-neutral.
- I propose four contractual milestones over approximately fourteen weeks, totalling USD $2,500, structured so that every milestone ends in a demonstration and a GO / NO GO decision.
- The architecture will be designed once, for the ten-year platform, and the Foundation Release will occupy only part of it. Future milestones should add files, not edit the core.
- Every account, repository and credential will be created in the Institute's name from the first day. I will hold delegated access, never sole access.
- Documentation will be a gated acceptance condition at every milestone, verified at the end by a cold-start deployment test performed on a clean environment using only the written documentation.

The Brief states that the Institute is seeking a long-term technology partner rather than a contractor. I have written this proposal on that basis: it contains recommendations the Institute did not ask for, and two places where I respectfully suggest doing less than the Brief proposes. I would rather be selected for how I think than for what I am willing to agree to.

2. Understanding of the Institute's Objective

The Collaborator Portal is not, in the ordinary sense, a networking product. It is the operational instrument of a curation process. The Institute's value does not come from the software; it comes from the judgement exercised about who is admitted. The software exists to let that judgement be exercised repeatedly, consistently, and at a distance.

That reading leads to two obligations on the Foundation Release:

1. The Institute must be able to run admission end to end — invitation, referral, application, review, decision, registration — without a developer's involvement and without touching a database. If any part of that loop requires me, the Portal is not production-ready, whatever else it does.
2. The first ninety seconds must be right. A distinguished professional in Helsinki or São Paulo will judge the Institute by an invitation email and a login page. The Institute gets one first impression with exactly the people it most wants, and the technical detail that most threatens it is not the interface — it is email deliverability. This is addressed in Section 6.

Everything in the Brief's functional scope serves one of those two obligations, which is why I regard the scope as well drawn.

3. Recommended Technical Foundation

The Brief asks for portability, simplicity, and freedom from dependence on any single developer. Those requirements point in a consistent direction: choose conventional, widely-known technology and use it in conventional ways. Novelty is a liability in a platform intended to outlive its first developer.

Layer	Recommendation	Why this, for this project
Application framework	Laravel 12 (PHP 8.3) or Node.js 22 with Next.js 15 (TypeScript)	Both are established, conventional, exceptionally well documented, and fully supported in-house — I build production systems in each and would deliver either to the same standard. Both give the Institute a deep global hiring pool, first-party authentication, queueing and mail, and no proprietary dependency. My recommendation and the basis for choosing between them follow this table.
Database	PostgreSQL 16	Relational integrity for the admission record, JSONB for profile attributes that will evolve, and native full-text and trigram search — which means search never requires a separate service at any scale the Institute will realistically reach.
Interface	Server-rendered interface with Tailwind CSS (Blade and Livewire on Laravel; React Server Components on Next.js)	In either case, server-rendered rather than a separate single-page application — less to maintain, deploy and secure. The designer's visual language is implemented once, in one place, and the Portal renders as a natural continuation of the Institute website.
Email	Amazon SES or Postmark, on the Institute domain	Authenticated sending with SPF, DKIM and DMARC, plus per-message delivery logging. Configured in Milestone 1, not at launch. See Section 6, Risk 1.
Hosting	Managed VPS with managed PostgreSQL	Predictable cost, no proprietary platform lock-in, and reproducible locally. Approximately USD $35–55 per month, billed to the Institute directly.
Code & CI	GitHub organisation owned by the Institute	Automated tests, dependency and security scanning on every change. The repository is the Institute's from commit one, not transferred at the end.

Choosing between Laravel and Node.js

I have deliberately presented both rather than a single option, because this is the one decision in the project that is genuinely difficult to revisit later, and the Institute should make it consciously rather than inherit it from whichever developer it happens to select. Everything else in this proposal — the data model, the admission workflow, the permissions structure, the documentation standard, the milestone plan and the budget — is identical under either choice.

Consideration	Laravel (PHP)	Node.js with Next.js (TypeScript)
Continuity of maintenance	Strongest position. Highly convention-driven — a new developer finds things where the framework says they will be, which is exactly what the Institute needs if	Strong, but conventions vary more between teams, so more depends on the discipline of the original build. Mitigated here by the documentation standard in

Consideration	Laravel (PHP)	Node.js with Next.js (TypeScript)
	responsibility ever transfers.	Question 5.
Hiring pool	Very deep and stable worldwide, at a consistently lower cost for maintenance work.	The largest pool overall, and the one growing fastest, though concentrated toward product work rather than long-term maintenance.
Fit to Foundation Release	Very strong. Invitations, roles, queued mail, admin tooling and audit logging are close to first-party — less bespoke code exists, so less can decay.	Equally capable, with slightly more assembly required for administrative tooling and background jobs.
Future roadmap	Excellent for API expansion and integrations. A separate front end can still be added later without disturbing the core.	Advantageous if the Institute later wants a shared codebase across web and mobile, or richer interactive features in the Portal.
Website integration	Natural if activelongevityinstitute.org runs on WordPress or similar PHP hosting — one environment, one set of credentials.	Requires separate hosting alongside the existing website. Entirely workable, marginally more infrastructure to document.

My recommendation is Laravel, on one criterion only: the Institute has told me this is a ten-year institutional asset that must never depend on a single developer. Under that constraint, depth of hiring pool and strength of convention matter more than language preference or technical elegance, and Laravel is the stronger answer.

That said, if the Institute anticipates a mobile application, expects richer interactive features in the Portal sooner than the roadmap suggests, or already has a JavaScript relationship it intends to keep, Node.js with Next.js is the better choice and I will build it to precisely the same standard, at the same price and on the same schedule. I would ask only that we settle it together in the first week of Milestone 1, and I will record the decision and its reasoning as the first architecture decision record so that whoever reads the repository in five years understands why.

4. Responses to Section 15

Question 1 — Foundation Release

Yes, with one addition and three simplifications. The scope in the Brief is well judged: it is the smallest set of functionality that lets the Institute actually operate, which is the correct definition of a foundation.

The addition I recommend: a permanent admission record

Every invitation issued, referral received, application submitted, and decision taken should be written to an immutable record — what happened, who did it, and when. Not an analytics feature; an institutional one.

Two years from now the Institute will need to answer questions such as: how was this collaborator admitted, who referred them, and on what basis was that application declined. History cannot be reconstructed if it was never recorded. Building this at the start costs a few days; adding it later costs the same few days and still leaves the first two years missing. I recommend including it in the Foundation Release, and I have priced it there.

A related point on data protection

The Institute intends to admit collaborators internationally, which will include the European Union and the United Kingdom. That places the Portal within scope of GDPR from its first registration. The practical implications are modest if handled at the start — an explicit lawful basis and consent capture at registration, a privacy notice, data minimisation, and the ability to export or erase a collaborator's data on request — and awkward if handled later, because remediation means returning to precisely the professionals the Institute has just cultivated. I have included the technical implementation in Milestone 2 and recommend the Institute have the accompanying notice reviewed by counsel.

Question 2 — Scope Management

I recommend deferring four things. In each case the underlying data model is still built in the Foundation Release — only the interface is postponed, so nothing is redesigned later.

- **Platform activity monitoring.** Analytics require a population before they mean anything; with thirty collaborators, a dashboard measures noise. I propose the permanent admission record plus a weekly digest email to the Founder instead, and a proper activity view once there is activity to view.
- **The notification management interface.** Nine transactional emails do not justify an editing system, and a mis-edited template reaching an invited professional carries a reputational cost the feature does not repay. I recommend templates held in version control for the Foundation Release, with any changes made by request, and an administrative editor introduced in a later milestone.
- **Dedicated surfaces for the richer profile fields.** Publications, languages, and collaboration availability should be captured in the schema and the profile form from day one — but they do not need their own browse or filter interfaces yet. Capture the data early; build the views when there is data worth viewing.
- **Advanced and faceted directory search.** The Brief already defers this and I want to record explicit agreement. Below roughly two hundred collaborators, name and expertise search alongside a browsable directory covers essentially all real behaviour.

Equally important is what I would not defer, whatever pressure the budget creates: multi-factor authentication for administrators, the admission record, rehearsed backup restoration, and email authentication. Each is inexpensive now and disproportionately expensive, or impossible, to retrofit.

Question 3 — Long-Term Architecture

The Brief's own phrasing is the right standard, and I would restate it as a test: every item on the Institute's future roadmap should be reachable by adding code, not by editing the core. Five commitments make that true.

- **Model the domain, not the release.** Collaborator, Invitation, Referral, Application, AdmissionDecision, Organisation and ExpertiseArea are first-class entities from the beginning, even where the Foundation interface touches only part of them. In particular, expertise is a controlled vocabulary the Institute manages — not free text. That single decision determines whether search, grouping, matching and reporting are possible later, or whether they begin with a data-cleaning project.
- **Roles and permissions as data, never as booleans.** There is no is_admin flag anywhere in this system. Roles and permissions live in tables and are enforced through authorisation policies, so the Brief's future "additional authorized Institute officers", regional reviewers, or read-only staff become configuration, not migration.
- **One service layer, two consumers.** Admission logic lives in service classes, not in controllers. The web interface calls them today; an HTTP API calls the same classes tomorrow. When a mobile application, an integration, or a partner system arrives, the rules are not reimplemented and cannot drift out of agreement with each other.
- **Events and queued jobs from day one.** Significant actions emit domain events; notifications subscribe to those events rather than being invoked inline. A future notification, digest, webhook or integration is then a new subscriber — an additive change — rather than surgery on working code. Queued delivery also means a slow mail provider never delays a person's registration.
- **Additive migrations and feature flags.** Schema changes add; they do not rewrite. Capability ships behind flags, so the Institute can enable something when it is ready rather than when a deployment window allows.

Question 4 — Risk Assessment

The genuine risks to this project are, in order, operational rather than technical.

Risk	Why it matters here	How I propose to manage it
1. Email deliverability	An invitation in a spam folder is an invitation that was never sent — and the Institute will never know. This is the Portal's single largest operational risk because the invitation is the Institute's first contact with a curated professional.	Authenticated sending on the Institute domain with SPF, DKIM and DMARC configured in Milestone 1. Inbox placement tested against Gmail, Outlook and university mail systems before any real invitation is sent. Per-message delivery status visible to the administrator, and one-click reissue of any invitation.
2. Dependency on the visual identity handover	The Brief places the logo, typography, palette and page design with the Institute's website designer. If those arrive late, interface work stalls or is built twice.	The Portal is built against a theme layer of design tokens, so identity is applied at the end without rework. I ask only for an agreed delivery date at Milestone 1 and proceed on neutral placeholders until then. Late assets cost the Institute nothing.

Risk	Why it matters here	How I propose to manage it
3. Scope growth against a fixed budget	The most common cause of failure in fixed-price institutional work is not that requirements change — it is that they change quietly and the cost is absorbed by reducing quality elsewhere, invisibly.	Milestone scope agreed in writing before work begins. Any change is logged with its cost and schedule impact and decided by the Institute at the milestone boundary. Nothing is absorbed silently and nothing is refused outright.
4. An empty directory at launch	A community platform showing fifteen half-complete profiles tells a distinguished professional they arrived too early. This is a product risk, not a software defect, and no amount of engineering fixes it.	Sequenced admission in waves, with a directory visibility control built in Milestone 3. Set out in full as my answer to Question 7.
5. Continuity — dependence on one developer	The Institute raises this itself, and it is a fair concern about any single-developer engagement, including mine.	Institute-owned repository and infrastructure from the first day. Documentation as a milestone acceptance condition rather than an end-of-project task. The cold-start deployment test described under Question 5, which proves the claim rather than asserting it.
6. Cross-border data protection	International collaborators bring the Portal within GDPR scope from the first registration. Remediation later means returning to the professionals the Institute has just recruited.	Consent capture and a privacy notice at registration, data minimisation by default, export and erasure capability, and a deliberate hosting-region decision made at Milestone 1. Legal review of the notice recommended.
7. Attack surface of an invitation-only system	Closed platforms attract credential stuffing and account enumeration, and the membership list itself is confidential.	Single-use, signed, expiring invitation tokens. Rate limiting on all authentication endpoints. Error messages that do not reveal whether an address is registered. Mandatory multi-factor authentication for administrators. Automated dependency scanning on every change.

Question 5 — Documentation

Documentation is treated as a deliverable of each milestone, reviewed at the milestone gate alongside the software. It is not a task at the end, because documentation written at the end is written from memory.

- README — written so a competent developer can run the Portal locally in under thirty minutes from a clean machine.
- ARCHITECTURE.md — the shape of the system, the boundaries between its parts, and how a new capability is added.

- Architecture Decision Records — numbered, short, one per significant choice, recording what was decided, what was rejected, and why. The reasoning is the part that decays; code explains itself, intent does not.
- Database documentation — an entity-relationship diagram and annotated schema generated from the migrations, so it cannot drift from reality.
- Operations runbook — how to deploy, roll back, restore from backup, rotate credentials, and respond to an incident.
- Administrator guide — written for the Founder rather than for an engineer, in plain language, with screenshots.
- Changelog — a plain-language record of what each milestone delivered.

The cold-start test

At the final milestone I will provision a clean server and deploy the Portal from nothing, using only the repository and the written documentation — no local knowledge, no memory, no shortcuts — and record the session for the Institute. If the documentation is inadequate, that failure becomes visible and is corrected before handover.

I propose that this test be a formal acceptance condition of the final milestone. It is the only honest way to verify the Institute's portability requirement, and I would rather be held to it than be trusted about it.

Question 6 — Milestone Recommendations

Four milestones over approximately fourteen weeks, totalling USD $2,500. Each is sized to deliver something the Institute can see working, so that every GO / NO GO decision is made against evidence rather than a progress report.

Milestone	Objective and principal deliverables	Duration	Budget
M1 **Foundation & Identity**	Objective: establish the architecture the next ten years will be built on, and the identity layer everything depends on. Institute-owned repository, CI pipeline, staging and production environments. Complete data model and migrations for the full domain, not only Phase 1. Authentication, password management and reset, secure sessions. Multi-factor authentication for administrators. Role and permission framework. Email infrastructure with SPF, DKIM and DMARC verified. Backup configured and a restore performed. Initial architecture decision records and README. Demonstration: an administrator signs in with MFA on staging; a test message is shown arriving in Gmail and Outlook inboxes; a database restore is performed live.	3.5 weeks	**$600**
M2 **Admission System**	Objective: the Institute's core operational workflow, end to end. Invitation creation, single-use expiring links, tracking and acceptance, and registration on acceptance. Structured referral form for approved collaborators. Become a Collaborator application form on activelongevityinstitute.org. A single administrative review queue covering both referrals and applications, with approve, decline and issue-invitation actions. Permanent admission record. Transactional notifications for every admission event. Consent capture and privacy implementation. Demonstration: a complete lifecycle run live — website application, review, decision, invitation, registration, admitted collaborator.	4 weeks	**$600**
M3 **Profiles, Directory & Institute Identity**	Objective: what the collaborator actually experiences. Profile creation and editing including photograph handling, with completion prompting. Searchable collaborator directory with browse and pagination. Search by name and by area of expertise. Institute-managed expertise vocabulary. Directory visibility control supporting phased launch. Implementation of the designer's visual language across the Portal, and the Collaborator Login entry point on the Institute	3.5 weeks	**$600**

Milestone	Objective and principal deliverables	Duration	Budget
	website. Demonstration: a seeded directory reviewed together, search exercised live, and a side-by-side visual comparison against the Institute website.		
M4 **Production Readiness & Handover**	Objective: move from working to trusted. Security review and hardening, rate limiting, performance pass. Uptime and error monitoring with alerting. Complete documentation set. Administrator guide and a live training session with the Founder. The cold-start deployment test, recorded. Production launch and transition support. Thirty days of defect correction after launch, included. Demonstration: the Institute issues its first real invitation.	2 weeks	**$700**
Total	Foundation Release, production deployed, documented and handed over	**~14 weeks**	**$2,500**

Durations exclude the Institute's review windows at each gate; allowing for those, I would plan on three and a half to four months end to end, consistent with the Brief.

Each milestone is authorised and paid separately, following the GO decision on the one before it. This matches the Institute's stated cash-flow objective and means the Institute is never more than one milestone exposed.

A note on the budget, offered plainly

USD $2,500 is achievable for the scope defined in this document, and I have sized the milestones so that it is. It is achievable because the scope is disciplined — not because the work is small. The mechanism that protects it is the change process described in Section 6, Risk 3: additions are priced and decided by the Institute, never quietly absorbed by cutting testing, documentation or security. I would rather raise a cost of $300 at the time than deliver something weaker and say nothing.

Question 7 — The One Recommendation

The greatest risk to this Portal is not technical. It is an empty room.

A distinguished professional accepts an invitation, signs in, opens the directory, and finds fifteen profiles, half of them incomplete. Nothing has malfunctioned. But the impression formed is that they arrived too early — and that impression is made once, with exactly the people the Institute most wants to keep. They will not return to check whether it improved.

> *My recommendation: do not open the directory on the day the Portal is deployed. Treat the first hundred admissions as a curated cohort released in waves, not as a rollout.*

- Wave 0 — roughly fifteen to twenty collaborators personally known to the Founder, invited with an explicit request to complete their profile fully. The directory is not yet open to them, or is presented as a founding-collaborators view.
- Wave 1 — roughly forty further collaborators, admitted once enough complete and credible profiles exist that the directory is itself an argument for joining.
- Wave 2 onward — referrals and website applications, which by then arrive into a community that already exists.

Technically this costs almost nothing: a per-collaborator flag and a directory visibility control, both of which I have included in Milestone 3 at no additional cost. What it buys is that every collaborator's first view of the Portal is of a community rather than a waiting room.

The software determines whether the Portal works. The sequencing determines whether it is believed in. If the Institute adopts only one recommendation from this proposal, I would ask that it be this one — it applies regardless of which developer is selected.

5. Ownership, Access and Portability — Commitments

Section 10 and Section 11 of the Brief describe the Institute's requirements. Rather than restate them, I am recording the commitments I make in response.

- Every account is created in the Institute's name from the first day — domain, DNS, hosting, database, GitHub organisation, email service, object storage, monitoring. Nothing is created under my name and transferred later, because transfers are where ownership gets lost.
- I request delegated administrative access only. I will never hold sole credentials to any asset of this project.
- Credentials are held in a password manager vault owned and controlled by the Institute.
- No proprietary components, no licence-encumbered code, and no dependency on any service operated by me. Every dependency will be open source and commonly used.
- At the close of each milestone: updated source code, database schema, deployment and configuration documentation, technical documentation, README, and a plain-language change log — exactly as set out in Section 11 of the Brief.
- At any point, on request and without negotiation, I will hand over access, documentation and a transition briefing. A partnership that cannot be ended cleanly is not a partnership.

6. Working Method

- A written progress note each Monday, timed for the Chicago morning: what was completed, what is next, anything that needs a decision.
- A staging environment available to the Institute continuously from Milestone 1, so progress can be seen rather than described.
- Each milestone closes with a recorded walkthrough and a live session, followed by the Institute's acceptance, revision request, or GO / NO GO decision.
- Questions and concerns raised when they arise rather than banked for a milestone meeting. Response within one business day.
- Where I disagree with a request, I will say so once, clearly, with reasoning — and then implement the Institute's decision properly. Section 16 of the Brief asks for professional judgement; this is how I intend to exercise it.

7. Assumptions and Exclusions

Recorded so that there is no ambiguity later.

- The Institute provides the domain and DNS control, the visual identity assets described in Section 12 of the Brief, and copy for public-facing pages.
- The Collaborator Login integration covers a styled entry point and the application form on the existing Institute website. A rebuild or redesign of activelongevityinstitute.org itself is not included.
- Third-party running costs — hosting, email, storage, approximately USD $40–60 per month in total — are billed to the Institute's own accounts and are not part of the $2,500.
- Content, copywriting and collaborator onboarding materials are the Institute's.

- I will implement the technical requirements of data protection and provide a structural draft of the privacy notice; legal review of that notice by the Institute's counsel is recommended and is not included.
- Scope beyond the Foundation Release as defined in the Brief is handled through the change process rather than absorbed.

8. Closing

The Brief says the Institute is looking for a technology partner rather than a contractor, and that the Portal is intended as an institutional asset rather than a software project. I have taken both statements literally in preparing this proposal, which is why it recommends doing less in three places, spends its longest section on documentation and portability, and gives its final recommendation to a launch sequencing decision that is not, strictly, my responsibility.

I would be glad to build this. I would also be glad to have been useful to the Institute's thinking regardless of the outcome — the waves recommendation in particular is worth adopting whoever is selected.

I am available to discuss any part of this proposal, and to begin Milestone 1 within one week of a decision.

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